

Canals - a canal-building mod for Civilization VII

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Canals

A Civilization VII mod. Build a Canal on a narrow neck of land between two bodies of water. When it is finished the tile becomes water, drawn as a canal for its age, and ships sail straight through it.



Figure 1: A Cog sailing through a three-tile Exploration canal cut across the neck of land beside Roma

An Exploration-age cut of three Canals, joining Roma's estuary to the northern sea. The channel, the houses and wharfs along its banks and the moored boats are drawn by the mod; the Cog's route through it is the game's own.

The Canal is an ordinary building in the production list. Each age has its own Canal, unlocked by one of that age's techs, and a city can dig as many canals as it has sites for. The mod decides where a Canal may go, turns the tile into water when it completes, and dresses it. Everything else, the ship pathing, the tile's ownership, the citizen who works it and the yields it gives, is the game's own state.

What the player does

Pick the Canal in a city's production list. It is a building; each age has its own, unlocked by a tech, and a city can build any number of them, one per site:

Age	Unlocked by	Production cost
Antiquity	Engineering	300
Exploration	Shipbuilding	500
Modern	Industrialization	800

On the placement screen the city offers only the tiles a canal makes sense on. In Antiquity that is a land tile the city owns with water on at least two separate sides: a strip of land between two shores. Sea, lake and navigable river all

count, so a canal can join a river to the sea or two reaches of a river. Three coast tiles on one side and one on the other qualifies; a headland with all its water in one stretch does not.



Figure 2: The city before the cut: the neck of land between the estuary and the sea

The site before the first Canal is ordered. The neck of land between Roma's estuary and the sea is three tiles across, which the Exploration age allows.

Choose a tile. It becomes an urban district holding the Canal under construction, and the city builds it like any other building.

When the Canal completes, the tile turns to water and is drawn as a canal at once, dressed for the age it was dug in: a water channel cut through the land toward each shore it joins, boats moored where it meets open water, and along the banks an Antiquity harbour with its stone quay, or Exploration houses and a wharf lining the channel, or Modern buildings and a harbour. Any ship can now path through the tile between the two waters, in the same turn.



Figure 3: The finished three-tile cut, close up

The finished cut. Each Canal that opens beside an earlier one redraws both so their channels meet, and a longer cut reads as one waterway.

Longer canals

From the Exploration age a canal can be two tiles long: a second Canal is offered on the tile beside the first that continues it in a straight line, and nowhere else around it. That is the whole Exploration allowance: no third tile, no bend, no branch. A queued Canal counts as part of the run, so the second can be ordered while the first is still being dug; ships pass once both are finished.

In the Modern age a run may be up to five tiles long and may bend and branch, as long as every tile of it reaches open water through the run. A junction where a branch leaves the main cut is drawn as an open basin, with no houses on it.

Age	Run	Bends and branches
Antiquity	1 tile	no
Exploration	a straight line of up to 2 tiles	no
Modern	up to 5 tiles	yes

In Antiquity a canal is always a single tile: the tiles beside an existing or queued canal are not offered. In every age a tile beside a canal is judged as part of that canal's run; a canal's own water never qualifies a neighbouring tile as an isthmus by itself.



Figure 4: An Antiquity canal beside the city centre, with a Galley in it

An Antiquity canal dug from the city's own tile to the sea, with a Galley in the channel. A settlement centre counts as a canal end, so a city or town can open its centre to the water.

Which tiles qualify

- flat or hill land the city owns, within the city's build ring, not a navigable river tile;
- ship-water on at least two separate sides of the hex: sea, lake or navigable river, in any combination (the same sea on both sides counts), ice not counted; two different bodies of water around it also qualify, even where they meet in one stretch; the settlement's own centre tile counts as water here, so a city or town can dig a canal from its centre out to the sea or a river, and a longer run may start or end at the centre;
- from Exploration on, alternatively the tile that extends an existing or queued canal within the age's run rule: a straight two-tile line in Exploration, up to five tiles with bends and branches in Modern;
- no urban district on it, and no building other than a rural improvement, which is removed when the Canal is placed;
- no unit standing on it;
- not already a canal, dug or queued.

What it gives, and what it costs

The Canal, like other buildings, takes one citizen to build. When it completes that citizen moves onto the finished canal, which becomes a worked fishing tile of the city, and the Canal building stays on it as an ordinary building of the city with its own yields, by the age it was dug in. All of it is ordinary city yield, counted where the city counts everything else. If the city cannot place the citizen on the canal itself, the tile still becomes a rural tile of the city and the citizen is left for you to place.

Canal	Food	Gold	Ships through it per turn
Antiquity	2	2	1
Exploration	3	4	2
Modern	4	6	no limit

The ship limit holds your own move orders: a move whose path runs through a canal that has already taken its ships this turn is not sent, and the count starts again each turn. A ship already sitting in the canal is not counted. The game's AI plots its routes natively and is not held.

The cost is the production and the tile: the rural improvement on it, if any, is gone, and the tile is water from then on.



Figure 5: A Modern cut with an Ironclad in it

A Modern canal, dug from the ocean to a navigable river, with an Ironclad on its way through. Modern canals are dressed with the age's harbour and buildings.

Cliffs and locks

A canal can be cut through a tile with cliffs on its edges. The cliffs stop mattering the moment the canal opens: ships cross every edge of the tile. The rock faces themselves stay in the picture until the game is next loaded, so where a channel meets a cliff shore the canal is drawn stepping down through a lock: a stone chamber with gates across the channel, a gatehouse on one bank, a water wheel on the other, and water spilling over the cliff foot beyond the gates. This is cosmetic only.



Figure 6: A canal meeting a cliff shore, drawn as a lock

Where the channel meets a shore that was a cliff, the canal is drawn as a lock.

More canals



Figure 7: A Modern run of three Canals at Pârsa, with an Ironclad in the channel

A Modern run of three Canals at Pârsa, cut from the northern sea to the river mouth in the south-west, with an Ironclad in the channel. The tile tooltip lists the Canal and the Fishing Boat on the tile, and each tile's food and gold is counted in the city.



Figure 8: The same Modern run close up, the Ironclad passing the buildings along the banks

The same run close up: the Ironclad passing the Modern buildings along the banks, the channel opening onto the sea beyond the pines.



Figure 9: A single Exploration canal at Ostia, joining the ocean to the bay behind the coast

An Exploration-age Canal at Ostia, on distant lands, joining the ocean to the bay behind the coast. The ship is on the bay side, about to go through.



Figure 10: The same canal from further out, the ship through on the ocean side

The same canal from further out. The ship is through, on the ocean side, and the canal tile reads as one more water tile of Ostia.

Compatibility

- Adds three buildings and their tech unlocks, changes no base-game rows, replaces no base-game files.
- A save loads with the mod on or off. A canal already dug is ordinary map state: a water tile worked by the city. The channel, quay and boats are drawn by the mod while it is on; without it the tile is plain water.
- A game started before the mod was enabled gets the Canal when its unlocking tech is researched; if that tech was researched earlier, that age's Canal is not available in that game.
- Single player. In a network game the Canal is not offered.
- English only for now. The mod's text is a handful of strings.

Installation

1. Subscribe on the Steam Workshop, or download the zip from the latest release.
2. Unzip it so the `canals` folder sits in the Civilization VII Mods directory.
3. Enable **Canals** from Additional Content in-game.

How it works

Five steps, each watched in the game on 2026-09-25 (Civilization VII 1.5.0). The run that proved each one is in docs/verification-runs.md.

1. **Placement.** The Canal buildings carry no terrain rule of their own. The mod wraps the engine's placement query: for a Canal, the offered plots are the city's land tiles within its build ring that pass the rule above, and the

per-plot check refuses anything else with the mod's own message. The engine's verdict on *whether* a Canal may be built (locked, already queued) is kept; the mod decides only *where*.

2. **Commit.** The engine offers buildings only on tiles that already hold an urban district, so a Canal on a rural isthmus first gets one, created by script, after which the Canal is offered there and a real build order queues it. The plot is remembered in the save, so a reload mid-construction still finishes the job. If the engine does not take the build within a few seconds, the district is removed again and the plot bought back.
3. **Completion.** When the Canal completes, the handler retypes the tile to coast (the feature cleared first), removes the Canal and its urban district (a district's housing block would hide the canal), buys the released plot back for the city, frees the Canal's citizen as a pending point and places it on the new water tile with the city's own expand order (a rural district with fishing boats, worked by that citizen), and re-creates the Canal building on that rural district, where its yield rows count in the city at once.
4. **The passage.** A tile retyped to coast is pathable by ships immediately: a ship's path to the far side goes from around the land to through the tile in the same turn. The retype survives a save and reload. The tile's area id does not change until the age transition, which is why the placement rule resolves a canal tile's waters through its neighbours rather than its area.
5. **The look.** A retyped hex keeps its land mesh until the next load, so the canal is drawn by script from shipped meshes: one river channel piece per water side meeting at the hex centre, the age's houses and harbour along the first arm, moored boats where the canal meets open water or a junction, and lock pieces at any shore that was a cliff. Three looks per age, chosen by plot number so a canal keeps its look across reloads. The opened canals are kept in the save and redrawn on load and at the start of every turn.

A safety sweep on load and at the start of every turn re-checks every remembered site and every complete Canal on the map, so a missed completion event is caught on the next turn. A completed Canal that shares its tile with other buildings (an AI's, in an urban slot) or does not sit on a canal site is left alone as a building.

Status

Watched working end to end on Civilization VII 1.5.0, 2026-09-25, in games of all three ages: placement through the real production screen, the district and the build order, the completion handler opening the canal, the citizen placed and the yields counted in the city, a Galley, a Cog and an Ironclad each sailing through, the three-tile Exploration cut and the Modern trunk ordered in sequence through the real build path, the lock dressing at a cliff shore, and short AI soaks with the Canal unlocked for every player without a crash.

Not watched:

- A refusal by the per-turn ship limit.
- A Modern branch built through the real build path: the branch tile was offered, but the engine dropped that one build order with no reason code in the gallery run, and branching was otherwise proved by retyping the tiles directly.
- The second and third look of each age as whole compositions (their pieces were photographed individually).
- A canal opening on an AI player's completion. The handler runs for any owner, but no AI built one in the soaks, which were run before the Canal had yields.
- The age transition with a canal on the map. The engine recalculates areas then, which should only make the canal more native.

Known limits:

- The hex mesh: a retyped tile keeps its land mesh until the next load, and the mod's overlay stands in for it. After a reload the engine draws the tile as water itself.
- Multiplayer: the retype is a local call, so the mod does not offer the Canal in a network game.

Layout

<code>canals.modinfo</code>	<code>data and one script</code>
<code>data/canals.xml</code>	<code>the three Canal buildings: cost, citizen, yields, districts</code>
<code>data/canals-antiquity.xml</code>	<code>each age's tech unlock, loaded only while that age is in use</code>
<code>data/canals-exploration.xml</code>	
<code>data/canals-modern.xml</code>	
<code>ui/canals.js</code>	<code>the mod: placement, commit, completion, looks, locks,</code>

text/en_us/CanalsText.xml	ship limit, safety sweep
docs/verification-runs.md	name, description and the placement message
docs/steam-workshop-description.txt	every step and the run that proved it
docs/workshop-preview.png	the Workshop page text
gallery/	the mod icon and Workshop preview (rendered from workshop-preview.svg)
scripts/build_readme_pdf.sh	the screenshots used above
scripts/steam-changelog.mjs	builds README.pdf from this file (npm run readme:pdf)
release.sh	renders CHANGELOG.md into CHANGELOG.steam.txt for the Workshop
CHANGELOG.md	builds the release zip and the Workshop manifest into dist/
README.pdf	what changed in each version
	this file, typeset

In a running game, `globalThis.__canals.enabled = false` makes every hook pass straight through, and `globalThis.__canals.uninstall()` restores the engine's own methods; both are for troubleshooting.

The routes that were tried and set aside, the probes and the harness runs live in the author's working tree (`mod_ideas_tested/canals`), outside this folder.

Credits

Built by Tower.